CSC 350 Assignment 7
Spring 2025

Due: Tuesday, April 29 (before class)

General Instructions

- Scan or compile snapshots of your submission into a PDF and upload it to Canvas.
- Assignments are **due before class** on the scheduled due date; generally, late assignments will not be accepted.
- Unless specified, **justify your answers always**. If justifications are missing or do not show reasonably how your answers are derived, 0 points will be assigned regardless of the correctness of the answers.
- Write **neatly** in the event you write your answers on paper. **Especially, for table completion questions, if your answers are not given neatly in well-drawn tables, the answers won't be graded**.
- **This assignment must be done by yourself without any unauthorized assistance**. Write your full name and numeric university ID on the front page of your submission, along with the following pledge statement.

I pledge my honor that all parts of this assignment were done by me individually and without any unauthorized assistance or collaboration with others.

Q1. Find a regular language to describe each of the following regular expressions.

(a) $a + bc$

(b) $a + b^*$

(c) $ab^* + bc^*$

Q2. Find a regular expression to describe each of the following languages.

(a) $\{aa, ab, ac\}$.

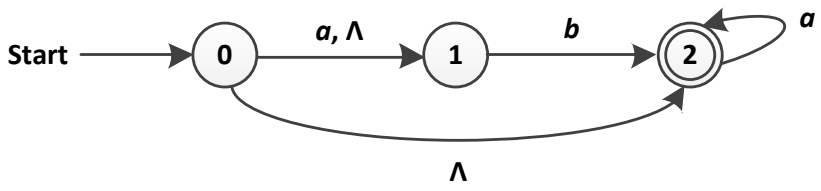
(b) $\{\Lambda, a, b, c, aa, bb, cc, ..., a^n, b^n, c^n, ...\}$.

(c) $\{a^m bc^n \mid m, n \in \mathbf{N}\}$

Q3. Find a regular expression over the alphabet $\{0, 1\}$ to describe the set of all binary numerals without leading zeros (except 0 itself).

So the language is the set $\{0, 1, 10, 11, 100, 101, 110, 111, ...\}$.

Q4. Suppose we are given the following NFA over the alphabet $\{a, b\}$:



(a) Find a regular expression for the language accepted by the NFA.

(b) Write down the transition table for the NFA.

(c) Use the algorithm to transform the NFA into a DFA.

(d) Draw a picture of the resulting DFA.

Q5. Given the following DFA table, use the algorithm to compute the minimum-state DFA. Answer parts (a), (b), and (c) as you proceed through the algorithm.

		a	b
start	0	1	2
	1	1	4
final	2	2	3
	3	3	4
final	4	4	2

(a) Find the set of equivalent pairs.

(b) Write down the states of the minimum-state DFA.

(c) Build the transition table for the minimum-state DFA.